

20250428EB_MN (Glo, imaging,IF)

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2025年4月28日 星期一

Use 2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A) or BP for differentiation and experiment.

Since the cells looked differently and not that healthy with T8 differentiation, I used more Na Pyruvate or BP to differentiate cells for test.

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年4月30日 星期三

T2 -> T3

2025年5月2日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年5月5日 星期一

T4 -> T5

2025年5月7日 星期三

T5 -> T6

2025年5月9日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年5月12日 星期一

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).

5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info			^
	A	B	
1	cell	passage	
2	WT11		

Table1			^
	A	B	
1	run1	volume (mL)	
2	trypsin	3	
3	HI-FBS/CTWM	6	
4	run2	*take some undissolved EB	
5	trypsin	2	
6	HI-FBS/CTWM	4	

Table3										✓
	A	B	C	D	E	F	G	H	I	
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume	
2	4.26	99	7	29.82	A	96	60	36	507	
3					B	24	250	16	939	
4					C,D	6	1972	12	the rest	

Plate A ^

	1	2	3	4	5	6
A				no virus		
B						
C			vLH92, 10uL in total			
D						

Plate B ^

	1	2	3	4	5	6
A	111, 12uL total					
B						
C	110, 12uL total					
D						

Plate C ^

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												
G												
H												

- 2025年5月13日 星期二
- Change T7 -> T7 if added virus
- 2025年5月14日 星期三
- T7-> T7 if not added virus
- 2025年5月15日 星期四

T7 -> T8. Plate B, C use T8 full supp with 0.5 mM pyruvate (no extra pyruvate)
Plate A: Use different T8 media for try.

Plate A – T8 media							^
	1	2	3	4	5	6	
A				T8 0.5mM pyruvate	T8 1.5mM pyruvate	BP T8 1.5mM pyruvate	
B							
C			T8 0.5mM pyruvate				
D			BP T8 1.5mM pyruvate				

2025年5月17日 星期六

1/2 T8 -> 1/2 T8 (use the same media as on 5/15)

2025年5月19日 星期一

1/2 T8 -> 1/2 T8

2025年5月21日 星期三

1/2 T8 -> 1/2 T8 (for plate B, C)

Imaging Plate A

media preparation						^
	A	B	C	D	E	
1	condition #	glucose conc. (mM)	volume	FCCP (1000x)	2CG (333x)	
2	1	2.5	2000			
3	2	0.15	2000			
4	3	0	2000			
5	4	0	2000	2	6	

Plate A – T8 media1 ^

	1	2	3	4	5	6
A				T8 0.5mM pyruvate	T8 1.5mM pyruvate	BP T8 1.5mM pyruvate
B						
C			1. 2.5mM	2. 0.15mM	3. 0mM	4. 0mM +FCCP +2DG
D			8.0mM +FCCP +2DG	7. 0mM	6. 0.15mM	5. 2.5mM

- Only image motor neuron this time (10 images/condition)

O.C. , 40X, zoom1.5, 600x900, 16-bit ^

	A	B	C	D
1	channel	bit	power (%)	exposure time
2	488	16	25	300
3	Dapi-ex- Green-em	16	20	500

NT: pre-treatment
no-glu: treat media as above
from 2:37-2:45 --> image the no glu groups at 3:41pm
(fileid == condition above)

IF Plate A

1. wash cells with cold PBS
2. fix cells with 4%PFA 10min, RT
3. PBS wash *1
4. ~~0.2% Triton X, 5-10min, RT~~ (forgot to do...)
5. 10% FBS in PBS blocking, 1h, RT
6. primary antibody in 10% FBS o/n at 4C

Table2								
	A	B	C	D	E	F	G	H
1		antibody	species	dilution			total volume	
2	C-32	Anti-MNX1 (HB9) Antibody	Rb	100	spinal motor neuron fate markers	row1	700	7
3	c-28	Purified anti-Tubulin β 3 (TUBB3) Antibody [TUJ1]	m	200	pan-neuronal markers	row1	700	3.5
4	121	VACHT	M	500	motor neuron marker	row2	700	1.4
5	C.13	MAP 2	Rb	500	pan-neuronal markers	row2	700	1.4
6	57	SOX2	Rb	1:500 (?)				

----2025/05/22-----

- 1. PBS wash 5 min *3
- 2. 2nd antibody in 10% FBS

2nd antibody1						
	A	B	C	D	E	F
1		antibody	fluoro	dilution	total volume	add volume
2	B	Anti-Rabbit IgG (H+L)	Alexa-488	1000	1200	1.2
3	D	Anti-Mouse IgG (H+L)	Alexa-647	1000	1200	1.2

- 3. PBS wash 5min *3
- 4. DAPI 5min
- 5. PBS wash 5min *3
- 6. mount

----2025/05/21-----

CellTiter Glo: Plate C

prepare ATP calibration							
	A	B	C	D	E	F	G
1	A	B	C	D	E	F	G
2	100mM	1mM	1uM	300nM	100nM	30nM	10nM
3	dilution factor from the previous one	100	1000		10		10
4					from C		fromE
5		1uL A in 99uL media	1uL B in 999uL	150uL in 350uL media	50uL C in 450uL media	150uL in 350uL media	50uL D in 450uL media

prepare media with different conc. of glucose1						
	A	B	C	D	E	F
1	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	1100	20	55	2000	36.4	1963.6
3	20	7.5	2.6666666667	1000	375.0	625.0
4	20	2.5	8	1600	200.0	1400.0
5	2.5	0.5	5	1000	200.0	800.0
6	2.5	0.15	16.6666666667	1000	60.0	940.0
7	2.5	0.05	50	1000	20.0	980.0
8	2.5	0.015	166.6666666667	1000	6.0	994.0

Plate C layout												^
	1	2	3	4	5	6	7	8	9	10	11	12
A	20						media blank					
B	7.5						C					
C	2.5						D					
D	0.5						E					
E	0.15						F					
F	0.05						G					
G	0.015											
H	0											

Table4				^
	A	B	C	
1		15:40	prepare Celltiter glo reagent	
2	-105	16:00	prepare media into a 96-well plate	
3	-75	16:07-16:10	add media to cells	
4	-23	17:00	prepare ATP standard and add samples	
5	-17	17:18	add celltiter glo reagent	
6	-12	17:21	put the plate into cytation 5 and start program to shake	
7	-10	17:23	wait	
8	0	17:33	measure	

2025年5月22日 星期四

IF Plate A (cont')

1. PBS wash 5 min *3
2. 2nd antibody in 10% FBS

2nd antibody							✓
	A	B	C	D	E	F	
1		antibody	fluoro	dilution	total volume	add volume	
2	B	Anti-Rabbit IgG (H+L)	Alexa-488	1000	1200	1.2	
3	D	Anti-Mouse IgG (H+L)	Alexa-647	1000	1200	1.2	

3. PBS wash 5min *3
4. DAPI 5min
5. PBS wash 5min *3
6. Check the staining and found I forgot to permeabilize the cells yesterday....
7. redo the Triton 0.2% treatment for 5min
8. 10% FBS blocking 1h, RT
9. 1st Antibody (600uL/row) o/n at 4C

Table7								
	A	B	C	D	E	F	G	H
1		antibody	species	dilution			total volume	
2	C-32	Anti-MNX1 (HB9) Antibody	Rb	100	spinal motor neuron fate markers	row1	600	6
3	c-28	Purified anti-Tubulin β 3 (TUBB3) Antibody [TUJ1]	m	200	pan-neuronal markers	row1	600	3
4	121	VACHT	M	500	motor neuron marker	row2	700	1.4
5	C.13	MAP 2	Rb	500	pan-neuronal markers	row2	700	1.4
6	57	SOX2	Rb	1:500 (?)				

2025年5月23日 星期五

IF Plate A (cont')

1. PBS wash 5 min *3
2. 2nd antibody in 10% FBS

2nd antibody2						
	A	B	C	D	E	F
1		antibody	fluoro	dilution	total volume	add volume
2	B	Anti-Rabbit IgG (H+L)	Alexa-488	1000	1200	1.2
3	D	Anti-Mouse IgG (H+L)	Alexa-647	1000	1200	1.2

3. PBS wash 5min *3
4. DAPI 5min
5. PBS wash 5min *3

Imaging

Plate B

vLH111 & 110

media preparation_SirDNA							
	A	B	C	D	E	F	G
1	condition #	glucose conc. (mM)	volume	FCCP (1000x)	2CG (333x)	SirDNA dilution	SirDNA volume
2	1	2.5	2000			5000	0.4
3	2	0.15	2000			5000	0.4
4	3	0	2000			5000	0.4
5	4	0	2000	2	6	5000	0.4

Table5						
	A	B	C	D	E	F
1			110		111	
2	number	condition	treatment time	image 1h	treatment time	image 1h
3	1	2.5mM	13:30	14:49	11:00	12:00
4	2	0.15mM	14:00	15:17	11:30	12:30
5	3	0mM	14:30	15:50	12:00	13:00
6	4	0mM+FCCP+2DG	15:35	17:11	13:00	14:00

#110 (1) kept out of focus -> delay

#110 (3) bubble passed through position 1 (12/30), 2 (17/30) of the second file.

#110 (4) kept having bubbles. I tried more than five time and there are still bubbles, so the imaging delayed a lot. But at the end, there are still some frames are out of focus because of the bubble...

out-of-focus frame ^

	A	B	C
1		110_4	110_3
2	1	24-30	11-14
3	2	10-13, 23-28	17-18
4	3	8-16, 21-23	N/A
5	4	10-14	N/A
6	5	N/A	N/A
7	6	N/A	N/A
8	7	N/A	N/A
9	8	N/A	N/A

Plate B condition ^

	1	2	3	4	5	6
A	1	2				
B	4	3				
C	1	2				
D	4	3				

O.C. , 60X, 600x900, 12-bit ^

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	640	12	30	300 ms	
3	561	12	10	100 ms	
4	488	12	50	1s	

8 positions/well

110: pre-activation: 3 loops, 30s interval; activation: 30 loops, 30s interval

111: pre-acitvation: 3 loops, 30s interval; activation: **25 loops**, 30s interval

Imaging: PlateA

fixed sample

Table6

^

	A	B	C
1		antibody	fluoro
2	rowA	Anti-MNX1 (HB9) Antibody	488
3		Purified anti-Tubulin β 3 (TUBB3) Antibody [TUJ1]	647
4	rowB	VACHT	488
5		MAP 2	647